Finals Math 2414 Litong Name ______________________ Test No, ________

Use the substitution formula to evaluate the integral.

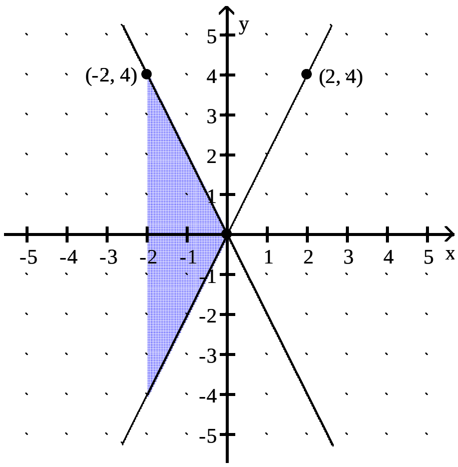
1) $\displaystyle\int_{\pi/6}^{\pi/2} \cot x \csc^5 x \, dx$

Find the area enclosed by the given curves.

2) $x = 2y^2 - 3$ and $x = y^2 + 6$

Answer each question appropriately.

3) Which of the following integrals, if any, calculates the area of the shaded region?



A) $\displaystyle\int_{-2}^{2} 4x \, dx$ B) $\displaystyle\int_{-2}^{0} 4x \, dx$

C) $\displaystyle\int_{-2}^{0} -4x \, dx$ D) $\displaystyle\int_{-4}^{4} -4x \, dx$

Find the volume of the solid generated by revolving the region about the given line.

4) The region in the first quadrant bounded above by the line $y = 3$, below by the curve $y = \sqrt{3x}$, and on the left by the y-axis, about the line $y = 3$

Use the shell method to find the volume of the solid generated by revolving the region bounded by the given curves and lines about the x-axis.

5) $y = 4|x|$, $y = 4$

Find the length of the curve.

6) $x = 6 \sin t + 6t$, $y = 6\cos t$, $0 \le t \le \pi$

Find the area of the surface generated by revolving the curves about the indicated axis.

7) $x = \sin t$, $y = 4 + \cos t$, $0 \le t \le 2\pi$; x-axis

Solve the problem.

8) It takes a force of 12,000 lb to compress a spring from its free height of 15 in. to its fully compressed height of 10 in. How much work does it take to compress the spring the first inch?

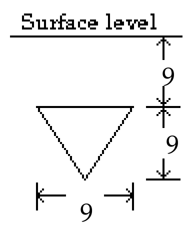
9) A fisherman is about to reel in a 16-lb fish located 14 ft directly below him. If the fishing line weighs 1 oz per foot, how much work will it take to reel in the fish? Round your answer to the nearest tenth, if necessary.

Find the center of mass of a thin plate of constant density covering the given region.

10) The region enclosed by the parabolas $y = -x^2 + 72$ and $y = x^2$

Find the fluid force exerted against the vertically submerged flat surface depicted in the diagram. Assume arbitrary units, and call the weight-density of the fluid w.

11)



Find the formula for df^{-1}/dx.

12) $f(x) = 27x^3$

Solve the problem.

13) If $f(x)$ is one-to-one, is $g(x) = f(-x)$ also one-to-one? Explain.

Find the derivative of y with respect to x, t, or θ, as appropriate.

14) $y = \dfrac{\ln x}{x^7}$

15) $y = \ln(\ln 5x)$

Evaluate the integral.

16) $\displaystyle\int \frac{2\,dx}{1 + 3x}$

17) $\int \frac{dx}{x(2 + 9 \ln x)}$

Use logarithmic differentiation to find the derivative of y.

18) $y = \sqrt{\frac{x}{x + 5}}$

Find the derivative of y with respect to x, t, or θ, as appropriate.

19) $y = \ln (7\theta e^{-\theta})$

20) $y = \sin e^{-\theta^6}$

Evaluate the integral.

21) $\int \frac{9e^{(9 \sin 5x)}}{\sec 5x} dx$

22) $\int \frac{e^{1/x}}{2x^2} dx$

Find the derivative of y with respect to the independent variable.

23) $y = (\ln 5\theta)^{\pi}$

Use logarithmic differentiation to find the derivative of y with respect to the independent variable.

24) $y = (\cos x)^x$

Find the derivative of y with respect to x.

25) $y = \tan^{-1} \frac{2x}{5}$

Evaluate the integral.

26) $\int \frac{7 - 5x}{\sqrt{64 - 49x^2}} dx$

27) $\int \frac{7 + 10x}{4 + 49x^2} dx$

28) $\int \frac{dx}{\sqrt{-x^2 - 6x - 8}}$

Solve the initial value problem.

29) $\frac{dy}{dx} = \frac{-6}{\sqrt{1 - x^2}}, \quad y(1) = -5$

Use l'Hôpital's rule to find the limit.

30) $\lim\limits_{x \to 0^+} \frac{\sin x}{x^4}$

Find the derivative of y.

31) $y = \sinh^2 9x$

Find the derivative of y with respect to the appropriate variable.

32) $y = 10 \tanh^{-1} (\cos x)$

Evaluate the integral.

33) $\int 2 \sinh (4x - \ln 3)\, dx$

34) $\int e^{2x} \cos 6x\, dx$

35) $\int x^4 \ln 8x\, dx$

36) $\int \csc^3 4t\, dt$

37) $\int_0^{\pi/2} \cos 8t \cos 7t\, dt$

Integrate the function.

38) $\int \frac{dx}{(x^2 + 4)^{3/2}}$

39) $\int \frac{dx}{x^2\sqrt{x^2 - 16}}, x > 4$

40) $\int \frac{1}{t^2\sqrt{9 - t^2}} dt$

Express the integrand as a sum of partial fractions and evaluate the integral.

41) $\displaystyle\int_3^7 \frac{3x\ dx}{(x-1)^3}$

Evaluate the integral by first performing long division on the integrand and then writing the proper fraction as a sum of partial fractions.

42) $\displaystyle\int \frac{7x^3 + 7x^2 + 8}{x^2 + x}dx$

Evaluate the improper integral or state that it is divergent.

43) $\displaystyle\int_0^\infty \frac{2dx}{36 + x^2}$

Evaluate the improper integral.

44) $\displaystyle\int_0^7 \frac{dx}{\sqrt{49 - x^2}}$

Find the limit of the sequence if it converges; otherwise indicate divergence.

45) $a_n = \dfrac{6 + (-1)^n}{6}$

Determine if the series converges or diverges; if the series converges, find its sum.

46) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1} \frac{4}{7^n}$

Determine whether the series converges.

47) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(\ln 6)^n}$

Find the slope of the polar curve at the indicated point.

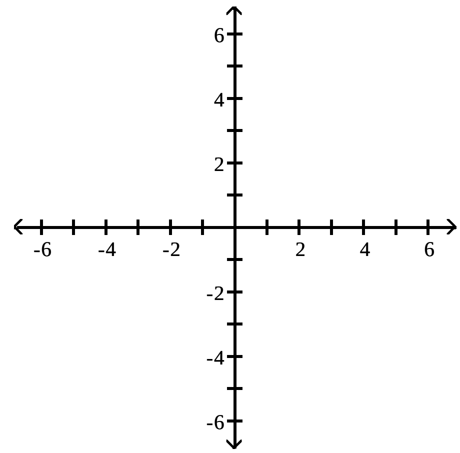
48) $r = 1 - \sin \theta,\ \theta = \dfrac{\pi}{3}$

Find the area of the specified region.

49) Inside one leaf of the four-leaved rose $r = 9 \sin 2\theta$

Graph the parabola or ellipse. Include the directrix that corresponds to the focus at the origin.

50) $r = \dfrac{12}{4 + 4 \cos \theta}$



1) $\frac{31}{5}$

2) 36

3) C

4) $\frac{9}{2}\pi$

5) $\frac{64}{3}\pi$

6) 24

7) $16\pi^2$

8) 1200 in • lb

9) 230.1 ft • lb

10) $\bar{x} = 0$, $\bar{y} = 36$

11) 486w

12) $\frac{1}{9x^{2/3}}$

13) g(x) is a reflection of f(x) across the y–axis. It will be one–to–one.

14) $\frac{1 - 7\ln x}{x^8}$

15) $\frac{1}{x \ln 5x}$

16) $\frac{2}{3} \ln|1 + 3x| + C$

17) $\frac{1}{9} \ln|2 + 9 \ln x| + C$

18) $\left(\frac{1}{2}\right)\sqrt{\frac{x}{x+5}}\left(\frac{1}{x} - \frac{1}{x+5}\right)$

19) $\frac{1}{\theta} - 1$

20) $(-6\theta^5 e^{-\theta^6}) \cos \ e^{-\theta^6}$

21) $\frac{1}{5} e^{(9 \sin 5x)} + C$

22) $-\frac{e^{1/x}}{2} + C$

23) $\frac{\pi}{\theta}(\ln 5\theta)^{\pi-1}$

24) $(\cos x)^x (\ln \cos x - x \tan x)$

25) $\frac{10}{4x^2 + 25}$

26) $\sin^{-1}\left(\frac{7}{8}x\right) + \frac{5}{49}\sqrt{64 - 49x^2} + C$

27) $\frac{1}{2} \tan^{-1}\left(\frac{7}{2}x\right) + \frac{5}{49} \ln|4 + 49x^2| + C$

28) $\sin^{-1}(x + 3) + C$

29) $y = 6 \cos^{-1} x - 5$
30) $-\infty$
31) $18 \sinh 9x \cosh 9x$
32) $\dfrac{-10}{\sin x}$
33) $\dfrac{1}{2} \cosh (4x - \ln 3) + C$
34) $\dfrac{e^{2x}}{40} [6 \sin 6x + 2 \cos 6x] + C$
35) $\dfrac{1}{5} x^5 \ln 8x - \dfrac{1}{25} x^5 + C$
36) $-\dfrac{\csc 4t \cot 4t}{8} - \dfrac{1}{8}\ln\left|\csc 4t + \cot 4t\right| + C$
37) $\dfrac{14}{15}$
38) $\dfrac{x}{4\sqrt{4 + x^2}} + C$
39) $\dfrac{1}{16} \dfrac{\sqrt{x^2 - 16}}{x} + C$
40) $-\dfrac{\sqrt{9 - t^2}}{9t} + C$
41) $\dfrac{4}{3}$
42) $\dfrac{7}{2}x^2 - 8 \ln\left| x + 1 \right| + 8 \ln\left| x \right| + C$
43) $\dfrac{\pi}{6}$
44) $\dfrac{\pi}{2}$
45) Diverges
46) Converges; $\dfrac{1}{2}$
47) converges
48) 1
49) $\dfrac{81\pi}{8}$

50)